Self Perform Electrical Design | BESS Electrical - CFD Menu of Services

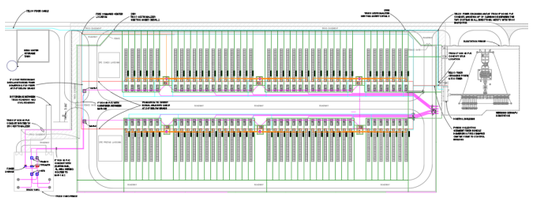
SPE DESIGN INVOLVEMENT, COORDINATION, & TIMELINE

McCARTHY

MEDIUM VOLTAGE/DIRECT CURRENT/AUX / FIBER UNDERGROUND AND INSTALL:

ELECTRICAL CONDUIT/CABLE COORDINATION AND ROUTING OF THE FOLLOW SCOPES:

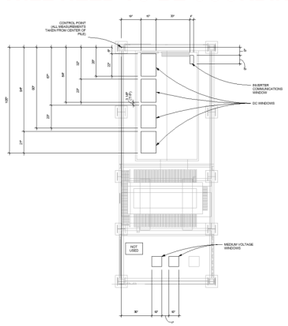
- MEDIUM VOLTAGE CABLE
- AUXILIARY CONDUCTOR ROUTING
- FIBER CABLE ROUTING
- DC FEEDER CABLE ROUTING
- FIRE SUPPRESSION SYSTEM ROUTING
- TRENCH WIDTHS
- BORING DETAILS
- SITE COORDINATION VS SWAILS, FENCES, PILES ETC...

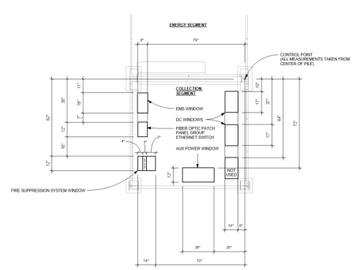


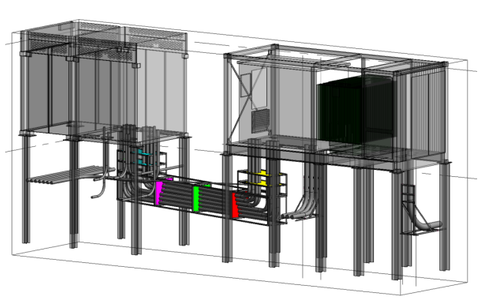
INVERTER / BESS SKID LAYOUT:

- 3D VIRTUAL MOCK UP OF SKID ROUGH IN INCLUDING:
- INVERTER SKID VS PILE COORDINATION
- SKID EQUIPMENT WINDOWS VS CONDUIT STUB UP COORDINATION
- CONDUIT QTY, SIZE, AND LAYOUT FOR MEDIUM VOLTAGE, DC FEEDERS, FIBER, AUX POWER, AND COMMUNICATIONS PER SKID
- STRUT SUPPORT LAYOUT INCLUDING LENGTHS AND DIMENSIONS

-NOTE: IF A FULLY COORDINATED MODEL IS DESIRED BY THE BESS TEAM. A MEETING TO DETERMINE THE LEVEL OF DETAIL FROM CFD WILL NEED TO BE COORDINATED WITH THE PM AND SUPERINTENDENT.

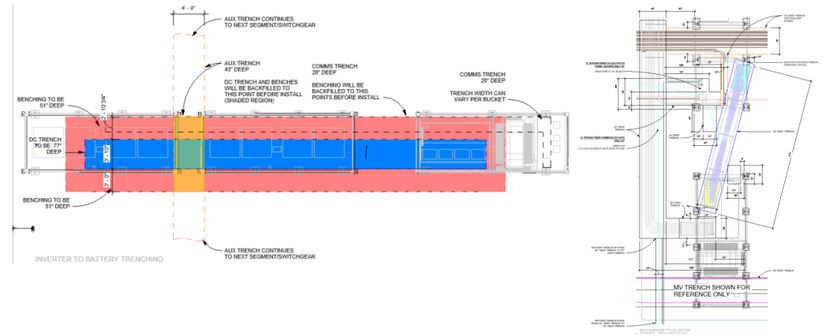






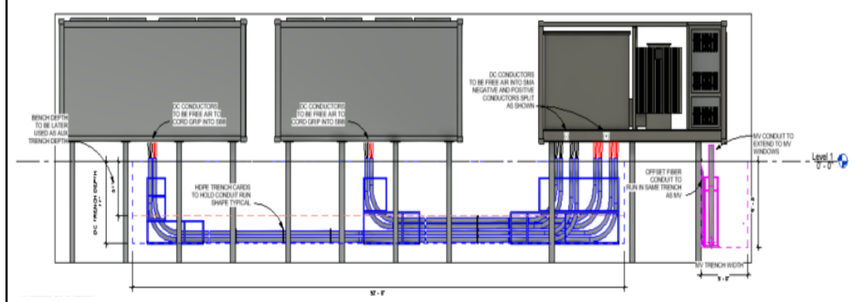
TRENCHING DETAILS:

- TRANSITION DETAILS
- PHOTO EXAMPLES OF COMPLETE INSTALL ON DELIEVERABLE
- GENERATION OF QUANTITIES OF MATERIALS FOR INSTALLATION



DC / MEDIUM VOLTAGE / FIBER / MISC. TERMINATIONS :

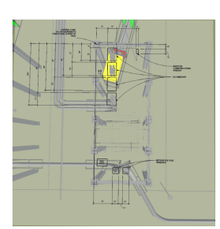
- CREATE DETAILS AND PHOTOS SHOWING HOW TO TERMINATE DIFFERENT EQUIPMENT AND/OR DEVICES



INTERACTIVE 3D REVIZTO MODEL

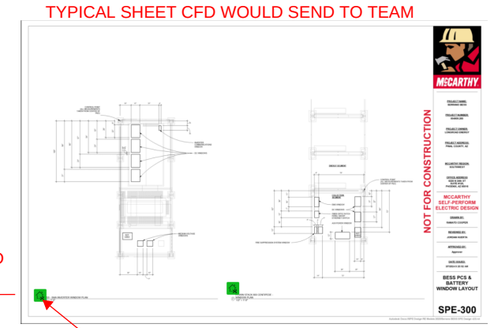
- 3D RENDERING OF CFD REVIT MODEL WHICH ALLOWS PROJECT TEAMS TO "FLY" AROUND THE MODEL TO REVIEW, CATCH CLASHES, TAKE MEASUREMENTS, ETC.
-2D DRAWINGS OVERLAY ONTO 3D FOR TEAMS TO REVIEW 3D MODEL AND DRAWINGS IN REAL TIME.

REVIZTO MODEL



2D AND 3D OVERLAY

TYPICAL SHEET CFD WOULD SEND TO TEAM



TEAMS CAN CLICK THIS ICON AND WILL OVERLAY 2D OVER 3D

PREFABRICATION DESIGN AND DETAILS:

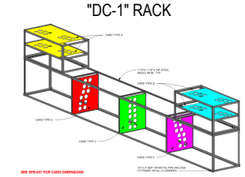
-FABRICATION DRAWINGS FOR CONDUIT RACKS
-DETAILS FOR CONDUIT INSTALLATION ON RACKS
-CONDUIT RACK INSTALLATION DRAWINGS
-LAYOUT POINTS FOR PLACING CONDUIT RACKS ON SITE
-OPTIMIZED SPE LAYDOWN AREA

PREFABRICATION RACK TYPES:(WILL VARY SLIGHTLY FROM JOB TO JOB)

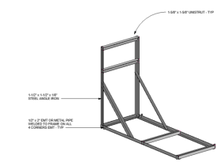
MV RACK-DOUBLE CONDUIT INTO SKID

MV RACK-SINGLE CONDUIT INTO SKID

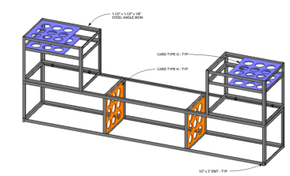
"DC-1" RACK



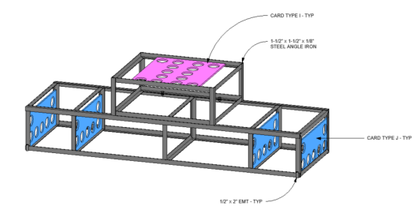
"DC-2" AUGMENTATION RACK



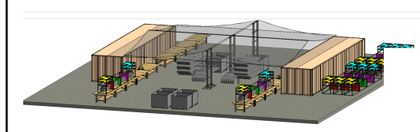
AUXILIARY POWER RACK - SWGR TO XFRMR



AUXILIARY POWER RACK-SWGR TO BESS



OPTIMIZED SPE LAYDOWN AREA



SUPERINTENDENT(S)
GENERAL FOREMEN(S)
PROJECT MANAGEMENT
PRECON/PROCUREMENT

COORDINATION
COMMUNICATION

CFD
SPE DESIGN

SEE REVERSE SIDE FOR TIMELINE / INVOLVEMENT INFORMATION →